AGRATAS GIGAFACTORY

Electrode Manufacturing

Standard Operating Procedures

CLARA Knowledge Base — Authorised SOP Corpus

Ingestion Reference: KB-ELEC-SOP-CORPUS-v1.0

SOPs Included	7 Operational SOPs + 1 Operator FAQ Document
Processes Covered	Mixing · Coating · Drying · Calendaring · Slitting · Shift Handover · Alarm Response
Access Tiers	Tier 1 (Operator) · Tier 2 (Supervisor) · Tier 3 (Production Manager)
Classification	INTERNAL — AUTHORISED PERSONNEL ONLY
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Electrode Slurry Mixing — Cathode & Anode

SOP ID	SOP-ELEC-MIX-001	Version	v2.1
Process Area	Electrode Mixing	Access Tier	Tier 1 / Tier 2
Effective Date	11/03/2026	Revision	Rev 3

1. Purpose

This SOP defines the controlled procedure for preparing cathode and anode electrode slurries used in lithium-ion battery manufacturing. Adherence is mandatory to ensure consistent slurry homogeneity, correct viscosity, and repeatable coating performance.

2. Scope

Applicable to all Line Operators and Shift Supervisors assigned to the Mixing Zone. Covers batch preparation from dry material weighing through final slurry quality check and transfer to coating.

3. Safety Requirements

SAFETY	Solvent vapours (NMP for cathode, water-based for anode) are present in the mixing zone. Ensure LEV (Local Exhaust Ventilation) is active before opening any containers. Wear chemical-resistant gloves, safety goggles, and lab coat at all times. NMP is a reproductive hazard — pregnant workers must not enter the mixing zone.

- Verify LEV extraction is operating before starting — check green indicator on panel.
- Wear nitrile gloves (min. 0.15 mm), safety goggles, and chemical-resistant lab coat.
- Keep emergency eyewash station clear and functional.
- Solvent containers must be sealed when not in active use.
- No open flames or ignition sources within 5 m of mixing area.

4. Materials & Equipment

Material / Equipment	Cathode (NMC)	Anode (Graphite)	Specification
Active Material	NMC811 powder	Artificial Graphite	Per BOM – batch card
Binder	PVDF (Kynar 761)	CMC + SBR	Battery grade
Solvent	NMP (anhydrous)	Deionised Water	H_2O < 30 ppm (NMP)
Conductive Additive	Carbon Black (C65)	Carbon Black (C65)	Per BOM
Mixing Vessel	Double planetary mixer	Double planetary mixer	Vessel clean, dry, certified

Material / Equipment	Cathode (NMC)	Anode (Graphite)	Specification
Viscometer	Brookfield DV2T	Brookfield DV2T	Calibrated, within cert date

5. Pre-Mixing Checks

ℹ NOTE All pre-mixing checks must be completed and initialled on the Batch Record before proceeding. Do not begin weighing until all checks pass.

1. Confirm work order number and batch card match. Record Batch ID on the mixing log.
2. Verify mixing vessel is clean and dry. Check vessel certification tag — must be within 7-day cleaning validity.
3. Confirm LEV is running: green LED on extraction panel is lit. Do NOT proceed if amber or red.
4. Check calibration status of balance (±0.1 g accuracy required). Confirm calibration sticker is current.
5. Check calibration status of viscometer. Confirm spindle number matches recipe card.
6. Confirm active material moisture content < 0.1% (check drying oven log for last pre-dry run).
7. Confirm NMP moisture content < 30 ppm. Check certificate of analysis on drum.
8. Record ambient temperature and humidity. Temperature must be 20–25 °C; Humidity < 30% RH. If out of range, notify Shift Supervisor before proceeding.

6. Mixing Procedure

6.1 Cathode Slurry (NMC)

⚡ CAUTION Add materials in the exact sequence below. Out-of-sequence addition causes agglomeration and will require the batch to be scrapped.

9. Weigh PVDF binder per batch card. Add to mixing vessel.
10. Add 60% of total NMP solvent. Mix at LOW speed (50 RPM) for 30 minutes until PVDF fully dissolved.
11. Add carbon black (C65). Mix at MEDIUM speed (500 RPM) for 45 minutes.
12. Add 50% of NMC active material. Mix at HIGH speed (1500 RPM) for 60 minutes.
13. Add remaining 50% NMC. Mix at HIGH speed (1500 RPM) for 60 minutes.
14. Add remaining 40% NMP to adjust viscosity. Mix at MEDIUM speed (800 RPM) for 20 minutes.
15. Apply vacuum (−0.08 MPa) for 30 minutes to degas slurry. Maintain low-speed mixing during degassing.
16. Proceed to Section 7 — Slurry Quality Check.

6.2 Anode Slurry (Graphite)

17. Dissolve CMC in deionised water (50% of total water). Mix at LOW speed (100 RPM) for 60 minutes.
18. Add carbon black. Mix at MEDIUM speed (500 RPM) for 30 minutes.
19. Add graphite active material in two equal portions, 30 minutes apart. Mix at HIGH speed (1200 RPM).

20. Add SBR binder. Mix at LOW speed (200 RPM) for 20 minutes. Do NOT use high speed with SBR — shear degrades binder.
21. Adjust final viscosity with remaining deionised water. Mix at MEDIUM speed for 15 minutes.
22. Proceed to Section 7 — Slurry Quality Check.

7. Slurry Quality Check

Parameter	Cathode Target	Anode Target	Action if Out of Spec
Viscosity (mPa·s)	4000–6000	2000–4000	Add solvent, mix 15 min, re-measure. Max 2 adjustments. If still OOS → reject batch.
Solid Content (%)	68–72%	45–50%	Reject batch if OOS. No adjustment.
Particle Size D50 (µm)	10–15	15–20	Re-mix high speed 30 min and re-test. If OOS → reject.
Magnetic Particle Content	< 0.1 ppm	< 0.1 ppm	Reject immediately. Notify QA and Supervisor.
Visual Appearance	Uniform, no lumps	Uniform, no lumps	Mix 30 min more and re-inspect. Lumps persist → reject.

CRITICAL Any batch failing magnetic particle content spec must be rejected immediately. Do not transfer to coating. Notify QA Lead and Shift Supervisor within 15 minutes. Retain the batch for investigation.

8. Slurry Transfer to Coating

23. Filter slurry through 100-mesh filter during transfer to coating vessel.
24. Record transfer volume and slurry batch ID on coating handover form.
25. Seal and label transfer vessel: Batch ID, slurry type, viscosity result, timestamp, operator ID.
26. Deliver to Coating Zone within 4 hours of batch completion. Record transfer timestamp.
27. Clean mixing vessel per Cleaning SOP-ELEC-CLEAN-003 immediately after transfer.

9. Alarms & Responses — Mixing Zone

Alarm Code	Description	Immediate Action
MIX-ALM-001	Mixer motor overtemperature	Stop mixer. Allow 15 min cool-down. Check for slurry hardening. If motor > 90 °C, call Maintenance.
MIX-ALM-002	LEV airflow low	STOP all solvent handling. Evacuate mixing area. Do not re-enter until LEV restored. Notify Maintenance and Supervisor.
MIX-ALM-003	Vessel vacuum loss	Pause degassing. Check pump and seals. If vacuum cannot be restored, degas with extended low-speed mix (60 min). Record deviation.

Alarm Code	Description	Immediate Action
MIX-ALM-004	Balance communication error	Do not proceed with weighing. Reboot balance. If persists, use backup and tag primary for maintenance.

Electrode Coating — Slot-Die Coating Process

SOP ID	SOP-ELEC-COAT-002	Version	v3.0
Process Area	Electrode Coating	Access Tier	Tier 1 / Tier 2
Effective Date	11/03/2026	Revision	Rev 2

1. Purpose

Defines the operating procedure for slot-die coating of electrode slurry onto aluminium (cathode) and copper (anode) current collector foil. Covers machine start-up, process parameter setting, in-process monitoring, and shutdown.

2. Safety Requirements

SAFETY

NMP solvent vapour is present around the coating machine. Drying oven operates up to 150 °C — risk of burns and solvent inhalation. PPE mandatory: heat-resistant gloves near oven exits, chemical-resistant gloves near slurry, safety goggles. Confirm oven door interlocks functional before start-up.

3. Pre-Coating Checks

28. Verify slurry batch ID and viscosity match coating work order. Slurry age must be < 4 hours. If > 4 hours, re-measure viscosity before proceeding.
29. Check foil reel: correct type (Al 16 µm ±1 µm for cathode; Cu 8 µm ±0.5 µm for anode), no visible defects (wrinkles, holes, scratches).
30. Inspect slot die: clean lips, no dried slurry. Die gap set per recipe. Die bolts torqued to 8 N·m.
31. Prime slurry pump: run until no air bubbles in return line (minimum 2 minutes).
32. Set oven temperatures per recipe. Allow oven to reach set point and stabilise for minimum 15 minutes.
33. Run 0.5 m trial coat. Check coating weight and appearance before production run.

4. HMI Process Parameters — Coating

Parameter	Cathode (NMC)	Anode (Graphite)	Alarm / Stop Limit
Coating Speed (m/min)	20–30	25–35	< 18 or > 38: stop line
Die Gap (µm)	150–200	180–220	±15 µm from target: adjust
Wet Coating Weight (g/m²)	280–320	200–240	±10%: stop, recheck die
Oven Zone 1 Temp (°C)	80	60	±5 °C: log; ±10 °C: stop
Oven Zone 2 Temp (°C)	120	90	±5 °C: log; ±10 °C: stop
Oven Zone 3 Temp (°C)	140	110	±5 °C: log; ±10 °C: stop

Parameter	Cathode (NMC)	Anode (Graphite)	Alarm / Stop Limit
Web Tension (N)	30–50	20–40	< 20 or > 60: stop line
Residual NMP in dried coat	< 50 ppm	N/A	> 200 ppm: reduce speed 10%, re-check

5. In-Process Quality Checks

Perform checks at the frequency shown. Record all results on the Coating Run Record. Stop line if any check fails.

Check	Frequency	Method	Acceptable Result
Wet coating weight	Every 200 m	Inline beta gauge + manual patch sample	Within ±10% of target g/m²
Dry coating weight	Every 500 m	100 cm² patch, weigh, subtract foil weight	Per recipe ±5%
Coating width	Every 200 m	Visual + tape measure against width guides	±1 mm of target
Surface visual inspection	Continuous	Visual: pinholes, streaks, uncoated spots	Zero pinholes; no streaks > 20 mm
Residual NMP (cathode only)	Every 1000 m	IR sensor inline + random GC patch	< 50 ppm NMP

6. Alarms & Responses — Coating

Alarm Code	Description	Response
COAT-ALM-001	Coating weight OOS	Stop at safe point. Inspect die gap and pump pressure. Adjust gap 5 µm increments. Trial 50 m. Re-check weight.
COAT-ALM-002	Oven temp deviation > 10 °C	Stop coating line. Do not advance foil through oven. Check heater and controller. Notify Maintenance if fault > 10 min.
COAT-ALM-003	Web break	E-stop immediately. Isolate oven energy. Re-thread foil per Threading SOP. Check tension after re-thread.
COAT-ALM-004	Slurry pump pressure low	Check slurry level in vessel. If not empty, check pump seals and filter. Do not coat with intermittent pump pressure.
COAT-ALM-005	Pinholes > 5 per 100 m²	Stop line. Check slurry for air bubbles (degas). Inspect and clean die lip if partially blocked.

Electrode Drying — Inline & Vacuum Oven Process

SOP ID	SOP-ELEC-DRY-003	Version	v1.4
Process Area	Electrode Drying	Access Tier	Tier 1 / Tier 2
Effective Date	11/03/2026	Revision	Rev 1

1. Purpose

Defines the controlled drying procedure to remove residual solvent from coated electrodes and achieve required moisture levels for calendaring and cell assembly. Covers both inline drying and vacuum oven batch drying.

2. Safety Requirements

SAFETY	Oven surfaces reach up to 150 °C. Thermal burns are a serious risk. Allow 30-minute cool-down before any maintenance access to oven zones. NMP vapour is extracted via oven exhaust — verify exhaust fan is running before operating.

3. Inline Drying — Operating Checks

Inline drying occurs automatically within the coating machine oven. Zone temperature targets are set on the coating HMI — see SOP-ELEC-COAT-002 Table 1.

34. Before starting, verify all three oven zones are at set point. Confirm exhaust flow indicator is green.
35. Monitor NMP exhaust concentration meter during production. If reading approaches 25% LEL, stop line and increase ventilation.
36. Check residual solvent in dried electrode every 1000 m using inline IR sensor. Target: < 50 ppm NMP.
37. If residual NMP > 200 ppm: reduce line speed by 10% and re-check. If still > 200 ppm after speed reduction, stop line and inspect oven heaters.

4. Vacuum Oven Batch Drying Procedure

Use for: (a) additional drying pass when inline drying fails spec, or (b) electrode reels awaiting coating > 24 hours after slurry application.

38. Pre-heat vacuum oven to 120 °C. Allow temperature to stabilise for 30 minutes.
39. Load electrode reels onto mandrel holders. Do not stack reels — each reel requires its own holder.
40. Seal oven door. Draw vacuum to < 10 Pa. Record vacuum level at start.
41. Hold at 120 °C / < 10 Pa for minimum 12 hours (cathode) or 8 hours (anode).

42. After drying cycle, allow oven to cool to < 40 °C under vacuum before breaking vacuum. Do NOT break vacuum while oven temperature > 60 °C.
43. Transfer dried reels immediately to dry room (RH < 1%) or nitrogen-purged storage. Maximum exposure to ambient air: 30 minutes.
44. Measure residual moisture by Karl Fischer titration from each reel. Target: < 50 ppm H_2O (cathode); < 100 ppm H_2O (anode).

5. Drying Failure Response

Scenario	Immediate Action	Escalation
Residual NMP > 200 ppm after inline drying	Reduce speed 10%. Inspect oven heaters. Route reel to vacuum oven.	Notify Supervisor if > 500 ppm. Hold reel — do not pass to calendaring.
Vacuum oven fails to hold vacuum (leak > 50 Pa/hr)	Stop cycle. Tag oven out-of-service. Transfer load to backup oven.	Notify Maintenance. Do not use oven until seals inspected.
Karl Fischer result OOS	Hold reel. Return to vacuum oven for 4-hour additional drying. Re-test.	If still OOS after 2nd pass, notify QA Lead. May require scrap disposition.

Electrode Calendaring — Density & Thickness Control

SOP ID	SOP-ELEC-CAL-004	Version	v2.2
Process Area	Electrode Calendaring	Access Tier	Tier 1 / Tier 2
Effective Date	11/03/2026	Revision	Rev 2

1. Purpose

Covers the calendaring (compression rolling) process to compact dried electrode coating to target density and thickness. Calendaring directly affects cell capacity, internal resistance, and electrolyte wetting.

2. Safety Requirements

SAFETY

Calendaring rolls exert up to 200 kN of force. The roll nip point is a No-Reach zone during operation. All guarding must be in place and interlocks active. No loose clothing or jewellery near rolls. Use push sticks to guide foil — never use hands near nip points.

3. Pre-Calendaring Checks

45. Verify electrode reel has passed drying quality check. Check reel tag for moisture result and sign-off. Do NOT calendar undried or OOS moisture reels.
46. Set roll gap per recipe. Verify with feeler gauge on both operator side and drive side. Gap difference must be < 2 µm.
47. Set roll temperature: Cathode 80 °C, Anode 25 °C (room temperature). Allow heated roll to stabilise for 20 minutes.
48. Set line speed per recipe. Do not exceed 20 m/min until first 100 m quality check confirmed.
49. Thread leader foil through rolls to verify alignment before loading electrode reel.

4. Process Parameters — Calendaring

Parameter	Cathode Target	Anode Target	OOS Action
Electrode Thickness (µm)	120–130	140–155	Stop, adjust roll gap.
Electrode Density (g/cm³)	3.3–3.5	1.5–1.6	Adjust gap; re-check.
Calendaring Force (kN)	100–180	60–100	Reduce if above maximum.
Roll Temperature (°C)	80 ± 5	25 ± 5	Stop if deviation > ±10 °C.
Line Speed (m/min)	10–20	15–25	Max 20 m/min until QC confirmed.
Porosity (%)	30–35	38–42	QA hold; notify process engineer.

5. In-Process Checks

50. Every 200 m: Cut 100 cm² sample. Measure thickness at 5 points (4 corners + centre). All must be within ±3 µm of target.
51. Every 500 m: Calculate electrode density from weight and thickness. Record on Calendaring Log.
52. Continuous visual check: Watch for foil wrinkles, coating cracking, or delamination. Any of these: stop line immediately.
53. If delaminating or cracking: reduce force by 10 kN and re-check. Continues → stop and notify Shift Supervisor. Likely cause: under-dried electrode or excessive force.

6. Alarms — Calendaring

Alarm Code	Description	Response
CAL-ALM-001	Roll gap deviation > 5 µm	Stop line. Re-zero gap sensor. Verify with feeler gauge. Recalibrate if required.
CAL-ALM-002	Calendaring force > 200 kN	Stop immediately. Reduce gap setting by 5 µm. Re-thread and restart at low speed.
CAL-ALM-003	Roll temperature OOS > ±10 °C	Stop foil advancement. Allow temperature to stabilise. If heater fault, call Maintenance.
CAL-ALM-004	Foil tension OOS	Stop line. Check unwind and rewind tension settings. Inspect foil for damage.

Electrode Slitting — Precision Width Cutting

SOP ID	SOP-ELEC-SLIT-005	Version	v1.8
Process Area	Electrode Slitting	Access Tier	Tier 1 / Tier 2
Effective Date	11/03/2026	Revision	Rev 2

1. Purpose

Defines the slitting procedure to cut parent electrode reels to the target width for cell assembly. Slit edge quality directly impacts cell safety (burr-induced short circuits) and overall yield.

2. Safety Requirements

SAFETY

Slitting blades are extremely sharp. Never touch blade edges. Use the blade-handling tool for all blade changes. Cut-resistant gloves (EN388 Level 4) mandatory during blade changes. Lock-out the slitter before any blade access.

3. Blade & Setup Checks

54. Confirm blade specification matches recipe (diameter, material, tooth profile). Check blade log for cumulative cut mileage. Replace blade set if cycles exceed 50,000 m.
55. Set blade spacing per recipe width (±0.1 mm tolerance). Verify with calibrated width gauge.
56. Check blade overlap (crush/shear setting): 0.1–0.3 mm crush for cathode; 0.1–0.2 mm crush for anode foil.
57. Check take-up reel core size matches downstream cell winding specification.
58. Inspect edge guide alignment — foil edge must run within ±0.5 mm of guide.

4. Process Parameters — Slitting

Parameter	Cathode	Anode	Alarm Limit
Slit Width (mm)	Per recipe ±0.1 mm	Per recipe ±0.1 mm	> ±0.2 mm: stop and adjust
Slitting Speed (m/min)	50–80	50–80	Do not exceed 100 m/min
Edge Burr Height (µm)	< 5	< 5	> 10 µm: change blades immediately
Foil Tension (N)	15–25	10–20	< 10 or > 30: stop
Edge Dust / Particles	None visible	None visible	Any: clean, inspect blade

5. In-Process Quality Checks

59. Every 500 m: Measure slit width at three points across reel width. Record on Slitting Log.

60. Every 500 m: Check edge burr using microscope or profilometer. Burr height must be < 5 µm.
61. Continuous: Watch for coating delamination at slit edge. Flaking: stop line and check blade condition.
62. At reel completion: Record slit reel ID, width measurements, burr check result, and blade cumulative mileage.

6. Slit Reel Labelling & Transfer

63. Each completed slit reel must be labelled: Reel ID, parent reel ID, electrode type, width, QC pass, date, operator ID.
64. Rejected reels (OOS width or burr): label clearly 'HOLD — QC REVIEW'. Move to designated hold area immediately. Do not mix with conforming reels.
65. Conforming reels transferred to cell assembly in sealed packaging within 8 hours to prevent moisture uptake.

Shift Handover — Electrode Manufacturing

SOP ID	SOP-ELEC-HANDOVER-006	Version	v1.2
Process Area	All Electrode Zones	Access Tier	Tier 1 / Tier 2
Effective Date	11/03/2026	Revision	Rev 1

1. Purpose

Defines the structured shift handover process for the electrode manufacturing area to ensure operational continuity, clear communication of status, and accurate transfer of accountability between outgoing and incoming shifts.

2. Outgoing Shift — Responsibilities

2.1 — 30 Minutes Before Shift End

66. Complete all active in-process quality checks. Record results on run records before handover.
67. Bring current production order to a safe stopping point. Do not initiate a new batch in the last 30 minutes unless instructed by Supervisor.
68. Note all active alarms, deviations, or quality holds: alarm code, time raised, action taken, current status.
69. Record consumable levels: slurry volume remaining, foil reel length remaining, blade mileage.

2.2 — At Shift Handover

70. Walk the incoming operator through the line. Point out any physical issues (unusual noise, smell, leaks).
71. Verbally communicate: current production order status, active quality holds, equipment restrictions.
72. Complete the Shift Handover Record on the MES system. All fields mandatory.
73. Do not leave the zone until the incoming operator has acknowledged handover verbally and signed the handover record.

3. Incoming Shift — Responsibilities

74. Arrive at zone 10 minutes before shift start. Review the MES Shift Handover Record before accepting handover.
75. Check all open alarms and quality holds listed in the handover record.
76. Physically inspect the line: equipment status lights, visible leaks, consumable levels.
77. Sign the handover record to accept accountability.
78. If you have concerns about the handover state, notify the Shift Supervisor before accepting handover.

4. Shift Handover Record — Required Fields

Field	Required Information
Production Order Number	Active order ID at time of handover
Current Reel / Batch ID	Most recent reel or batch in progress
Production Output	Metres coated / slit / calendared this shift
Open Alarms	All alarm codes + status (active / cleared / deferred)
Quality Holds	Reel/batch IDs on hold + reason + QA contact
Equipment Restrictions	Machines tagged out, restricted speed, under maintenance
Consumable Status	Slurry remaining, foil reel length, blade mileage
Pending Actions	Tasks carried forward for incoming shift
Outgoing Operator ID + Signature	Mandatory — handover invalid without signature
Incoming Operator ID + Signature	Mandatory — acceptance invalid without signature

General Alarm Response & Escalation — Electrode Zone

SOP ID	SOP-ELEC-ALARM-007	Version	v1.0
Process Area	All Electrode Zones	Access Tier	Tier 1 / Tier 2 / Tier 3
Effective Date	11/03/2026	Revision	Rev 1

1. Purpose

Defines the standard response framework for all process alarms in the electrode manufacturing area. All operators must follow this procedure when any alarm is triggered. Machine-specific responses are in the respective machine SOPs.

2. Alarm Priority Classification

Priority	Colour / Sound	Definition	Response Time Required
P1 — Critical	RED / Continuous horn	Immediate safety or quality risk. Production must stop.	Acknowledge within 30 seconds.
P2 — High	AMBER / Pulsed horn	Process deviation requiring action. May impact quality.	Acknowledge < 2 min. Resolve < 10 min.
P3 — Medium	YELLOW / Audible beep	Advisory. Monitor closely. Production may continue.	Acknowledge < 5 min. Log and monitor.
P4 — Low	BLUE / Visual only	Informational. No immediate action required.	Acknowledge < 15 min. Log.

3. General Alarm Response Procedure

79. ACKNOWLEDGE the alarm on the HMI within the required response time for its priority class.
80. IDENTIFY the alarm: read alarm code, description, and affected zone on HMI alarm screen.
81. ASSESS: Is production safe to continue? If any doubt, STOP the line. When in doubt, stop.
82. CONSULT the machine-specific SOP for the alarm code (MIX-001, COAT-002, DRY-003, CAL-004, or SLIT-005).
83. TAKE the action specified in the machine SOP. Do not deviate from the specified response.
84. RECORD the alarm on the Alarm Log: alarm code, time, action taken, resolved time, operator ID.
85. ESCALATE if alarm cannot be resolved within specified response time — see Section 4.

4. Escalation Matrix

Condition	Escalate To	Time Limit	Method
P1 alarm — any duration	Shift Supervisor	< 1 minute	Verbal + Andon pull

Condition	Escalate To	Time Limit	Method
P2 alarm not resolved in 10 min	Shift Supervisor	10 minutes	Verbal + MES notification
Recurring P2/P3 (> 3× per shift)	Shift Supervisor + Maintenance	Before end of shift	MES + verbal
Quality hold raised — any alarm	Supervisor + QA Lead	< 15 minutes	MES hold + verbal
Equipment fault requiring maintenance	Maintenance team	Immediate (P1); 30 min (P2)	CMMS work order + verbal
Alarm cause unknown / cannot resolve	Production Manager	Before production restart	MES + verbal escalation

CRITICAL Never clear (reset) a P1 alarm without physically resolving its root cause. Clearing an active alarm without resolution is a serious non-conformance and will be investigated as a quality and safety event.

Operator FAQ — Electrode Manufacturing Knowledge Hub

SOP ID	KB-ELEC-FAQ-001	Version	v1.0
Process Area	All Electrode Zones	Access Tier	Tier 1 / Tier 2 / Tier 3
Effective Date	11/03/2026	Revision	Rev 1

This FAQ provides quick answers to common operator questions. All answers link to the authoritative SOP. When in doubt, always refer to the full SOP.

Mixing

Q: What do I do if the slurry viscosity is too high after mixing?

A: Add small increments of solvent (NMP for cathode; deionised water for anode). Mix at medium speed for 15 minutes after each addition, then re-measure viscosity. Maximum 2 adjustment attempts. If still OOS after 2 attempts, reject the batch and notify your Shift Supervisor. Reference: SOP-ELEC-MIX-001 §7.

Q: Can I start mixing if ambient humidity is above 30% RH?

A: No. Mixing must not start if humidity exceeds 30% RH. NMP absorbs moisture which degrades slurry quality. Log the environmental condition, notify your Shift Supervisor, and wait until humidity is within range. Reference: SOP-ELEC-MIX-001 §5, Step 8.

Q: The LEV green light is not lit. What do I do?

A: Do NOT open any solvent containers or start mixing. This is a P1 safety condition. Notify your Shift Supervisor immediately and activate the Andon. Do not re-enter the mixing zone until LEV is confirmed operational. Reference: SOP-ELEC-MIX-001 §3 and SOP-ELEC-ALARM-007 §2.

Q: How do I know if the mixing vessel is clean enough to use?

A: Check the vessel certification tag on the vessel. The cleaning date must be within the 7-day cleaning validity window. If the tag is absent, expired, or unclear, treat the vessel as unclean and route for cleaning before use. Reference: SOP-ELEC-MIX-001 §5, Step 2.

Coating

Q: How often do I check coating weight during a production run?

A: Check wet coating weight every 200 m using the inline beta gauge plus a manual patch sample. Check dry coating weight every 500 m by cutting a 100 cm² patch, weighing it, and subtracting the foil weight. Record all results on the Coating Run Record. Reference: SOP-ELEC-COAT-002 §5.

Q: I can see pinholes in the coating. What should I do?

A: Stop the line if you observe more than 5 pinholes per 100 m². Check the slurry for air bubbles — it may need to be re-degassed. Inspect and clean the die lip for partial blockage. This corresponds to alarm code COAT-ALM-005. Reference: SOP-ELEC-COAT-002 §6.

Q: How long can slurry wait before I coat with it?

A: Maximum 4 hours from slurry batch completion to start of coating. If the slurry is older than 4 hours, re-measure viscosity before proceeding. If viscosity is OOS, do not coat — reject the slurry and notify the Shift Supervisor. Reference: SOP-ELEC-COAT-002 §3, Step 1.

Q: What do I do if the oven temperature goes outside the alarm limit?

A: A deviation of ±5 °C from set point requires logging and monitoring. A deviation of ±10 °C requires stopping the coating line. Do not advance foil through the oven. Check the heater and temperature controller, and notify Maintenance if the fault persists for more than 10 minutes. Reference: SOP-ELEC-COAT-002 §6, COAT-ALM-002.

Drying

Q: What is the maximum residual NMP allowed in a dried cathode electrode?

A: Less than 50 ppm NMP. If the inline IR sensor reads above 200 ppm, reduce line speed by 10% and re-check. If residual NMP remains above 200 ppm after speed reduction, stop the line and inspect the oven heaters. Reference: SOP-ELEC-DRY-003 §3.

Q: How long must cathode reels dry in the vacuum oven?

A: Minimum 12 hours at 120 °C with vacuum maintained below 10 Pa. Anode reels require a minimum of 8 hours. Do not break the vacuum until the oven has cooled below 40 °C. Reference: SOP-ELEC-DRY-003 §4.

Q: How long can a dried reel be exposed to ambient air before it must be used?

A: Maximum 30 minutes exposure to ambient air after the vacuum oven drying cycle. After that, the reel must be stored in a dry room (RH < 1%) or nitrogen-purged storage. Reference: SOP-ELEC-DRY-003 §4, Step 6.

Calendaring

Q: The electrode coating is cracking at the rolls. What is the likely cause and what should I do?

A: Cracking is most commonly caused by an under-dried electrode or excessive calendaring force. Stop the line immediately. Reduce calendaring force by 10 kN and re-check. If cracking continues, notify your Shift Supervisor — the reel may need to be returned for additional drying. Reference: SOP-ELEC-CAL-004 §5, Step 4.

Q: How do I verify the roll gap is equal on both sides?

A: Use a feeler gauge on both the operator side and the drive side of the roll gap. The difference between the two sides must be less than 2 µm. If the difference is greater, adjust the drive-side screw until both sides are within 2 µm of each other. Reference: SOP-ELEC-CAL-004 §3, Step 2.

Q: What do I do if the calendaring force exceeds 200 kN?

A: This is alarm CAL-ALM-002. Stop the line immediately. Reduce the roll gap setting by 5 µm. Re-thread the foil and restart at low speed. Do not continue calendaring with force above 200 kN. Reference: SOP-ELEC-CAL-004 §6.

Slitting

Q: When do I need to change the slitting blades?

A: Replace the blade set when cumulative cut length reaches 50,000 m, or immediately if edge burr height exceeds 10 µm at any point during the shift. Check the blade log for current mileage at the start of each shift. Reference: SOP-ELEC-SLIT-005 §3, Step 1.

Q: What should I do with a slit reel that has failed the width check?

A: Label the reel clearly as 'HOLD — QC REVIEW'. Move it to the designated hold area immediately. Do not mix it with conforming reels. Record the reel ID, width measurement, and time on the Slitting Log, then notify your Shift Supervisor. Reference: SOP-ELEC-SLIT-005 §6, Step 2.

Q: How soon must conforming slit reels be transferred to cell assembly?

A: Within 8 hours of completion, in sealed packaging. This prevents moisture uptake which could degrade electrode performance. Reference: SOP-ELEC-SLIT-005 §6, Step 3.

Shift Handover

Q: Can I leave the zone before the incoming operator signs the handover record?

A: No. You must remain in the zone until the incoming operator has acknowledged the handover verbally AND signed the MES Shift Handover Record. Leaving without a completed handover is a non-conformance. Reference: SOP-ELEC-HANDOVER-006 §2.2, Step 4.

Q: What fields must I complete on the Shift Handover Record?

A: All fields are mandatory: Production Order Number, Current Reel/Batch ID, Production Output (metres), Open Alarms (codes + status), Quality Holds (reel IDs + reason), Equipment Restrictions, Consumable Status, Pending Actions, and both operator signatures. An incomplete handover record must be flagged to the Shift Supervisor. Reference: SOP-ELEC-HANDOVER-006 §4.

General Safety

Q: What is the emergency stop (E-stop) procedure?

A: Press the nearest red mushroom-head E-stop button. This immediately de-energises the machine. Shout 'E-STOP ACTIVATED' to alert nearby workers. Do NOT restart the machine after an E-stop without first investigating and resolving the cause, and obtaining Shift Supervisor authorisation. Notify the Shift Supervisor of every E-stop event.

Q: I can smell strong solvent but no alarm has triggered. What should I do?

A: Evacuate the area immediately and activate the nearest Andon. Do not wait for an alarm. Notify the Shift Supervisor verbally. This may indicate a LEV failure or solvent spill before the sensor has detected it. Re-entry only with Supervisor and EHS authorisation. Reference: SOP-ELEC-ALARM-007 §3, Step 3 — when in doubt, stop.

Q: If a quality hold is raised, who do I notify and how quickly?

A: Notify your Shift Supervisor and QA Lead within 15 minutes of raising the hold. Use MES hold notification plus a verbal communication. Do not release a quality hold without QA Lead sign-off. Reference: SOP-ELEC-ALARM-007 §4.

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